Vignette for peakTrams R-package

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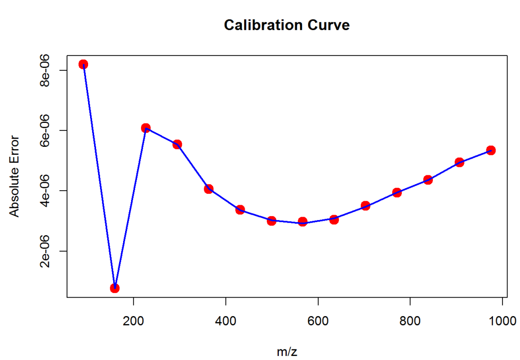
```
Step 1.Specify the paths of the raw files
path_HILIC<-list.files(paste(find.package(package="peakTrams"),"data",sep="/"),
              pattern = c("Sample_HILIC",".mzXML"), full.names = TRUE)
path_RP<-list.files(paste(find.package(package="peakTrams"),"data",sep="/"),
              pattern = c("Sample_RP",".mzXML"), full.names = TRUE)
path_Blank_HILIC<-list.files(paste(find.package(package="peakTrams"),"data",sep="/"),
              pattern = c("Blank_HILIC",".mzXML"), full.names = TRUE)
path_Blank_RP<-list.files(paste(find.package(package="peakTrams"),"data",sep="/"),
              pattern = c("Blank_RP",".mzXML"), full.names = TRUE)
```

```
Step 2.Load libraries needed
library("peakTrams")
```

```
Step 3.Read, calibrate mzXML and restrict processing area
dir.create(c("output"),showWarnings=T)
setwd(as.character(paste(getwd(),c("output"), sep="/")))

sample_RP<- calibration(mzXML=read.mzXML(filename=path_RP),
              calibration_region=c(0.08,0.26),
              calibrant_substance="Na_Formate_pos",int_thres=1000)
blank_RP<- calibration(read.mzXML(filename=path_Blank_RP),
              calibration_region=c(0.08,0.26),
              calibrant_substance="Na_Formate_pos",int_thres=1000)
sample_HILIC<- calibration(read.mzXML(path_HILIC),
              calibration_region=c(0.08,0.26),
              calibrant_substance="Na_Formate_pos",int_thres=1000)
blank_HILIC<- calibration(mzXML=read.mzXML(path_Blank_HILIC),
              calibration_region=c(0.08,0.26),
              calibrant_substance="Na_Formate_pos",int_thres=1000)
```



```
blank_HILIC<-removescans(mzXML=blank_HILIC,scansORtime=c("beginning",0.26),time=TRUE)
blank_HILIC<-removescans(mzXML=blank_HILIC,scansORtime=c(17,"end"),time=TRUE)
```

```
Beginning point was set at 16.954 because given ending retention time 17 corresponds to s
can at MS2 level
```

```
sample_HILIC<-removescans(mzXML=sample_HILIC,scansORtime=c("beginning",0.26),time=TRUE)
sample_HILIC<-removescans(mzXML=sample_HILIC,scansORtime=c(17,"end"),time=TRUE)
```

```
Beginning point was set at 16.9542 because given ending retention time 17 corresponds to
scan at MS2 level
```

```
sample_RP<-removescans(mzXML=sample_RP,scansORtime=c("beginning",0.26),time=TRUE)
blank_RP<-removescans(mzXML=blank_RP,scansORtime=c("beginning",0.26),time=TRUE)
```

```
Step 4.Scan-by-scan Subsraction
sub_HILIC<-subtract(sample=sample_HILIC,blank=blank_HILIC,mzdiff=0.01,filter=100)
```

```
Sample has 2008 scans from which 335 scans are in MS1 level and 1673 in MS2 level.
MS2 scans are disregarded during the subtraction procedure
Blank has 2008 scans from which 335 scans are in MS1 level and 1673 in MS2 level.
MS2 scans are disregarded during the subtraction procedure
|=================================================================| 100%
```

```
sub_RP<-subtract(sample=sample_RP,blank=blank_RP,mzdiff=0.01,filter=100)
```

```
Sample has 1825 scans from which 321 scans are in MS1 level and 1504 in MS2 level. MS2 sc
ans are disregarded
during the subtraction procedure
Blank has 1825 scans from which 394 scans are in MS1 level and 1431 in MS2 level. MS2 sca
ns are disregarded
during the subtraction procedure
|=================================================================| 100%
```

```
Step 5.Write the subtracted chromatograms
write.mzXML(sub_RP, paste("Subtracted_RP",".mzXML", sep=""), precision=c('64'))
```

```
write.mzXML(sub_HILIC, paste("Subtracted_HILIC",".mzXML", sep=""), precision=c('64'))
```

```
  Step 6.Peak picking and comparing of peaklists
  output<-compareRPHILIC(HILIC=paste(getwd(),"Subtracted_HILIC.mzXML",sep="/"),
                RP=paste(getwd(),"Subtracted_RP.mzXML",sep="/"),
                mzthr=0.01, doubleckeck=TRUE,
                ppm=17.6, peakwidth=c(14.34,50), prefilter=c(1,1000) , sn=10)
```

```
Detecting mass traces at 17.6 ppm ...
% finished: 0 10 20 30 40 50 60 70 80 90 100
1977 m/z ROI's.

Detecting chromatographic peaks ...
% finished: 0 10 20 30 40 50 60 70 80 90 100
954  Peaks.
|=================================================================| 100%
```

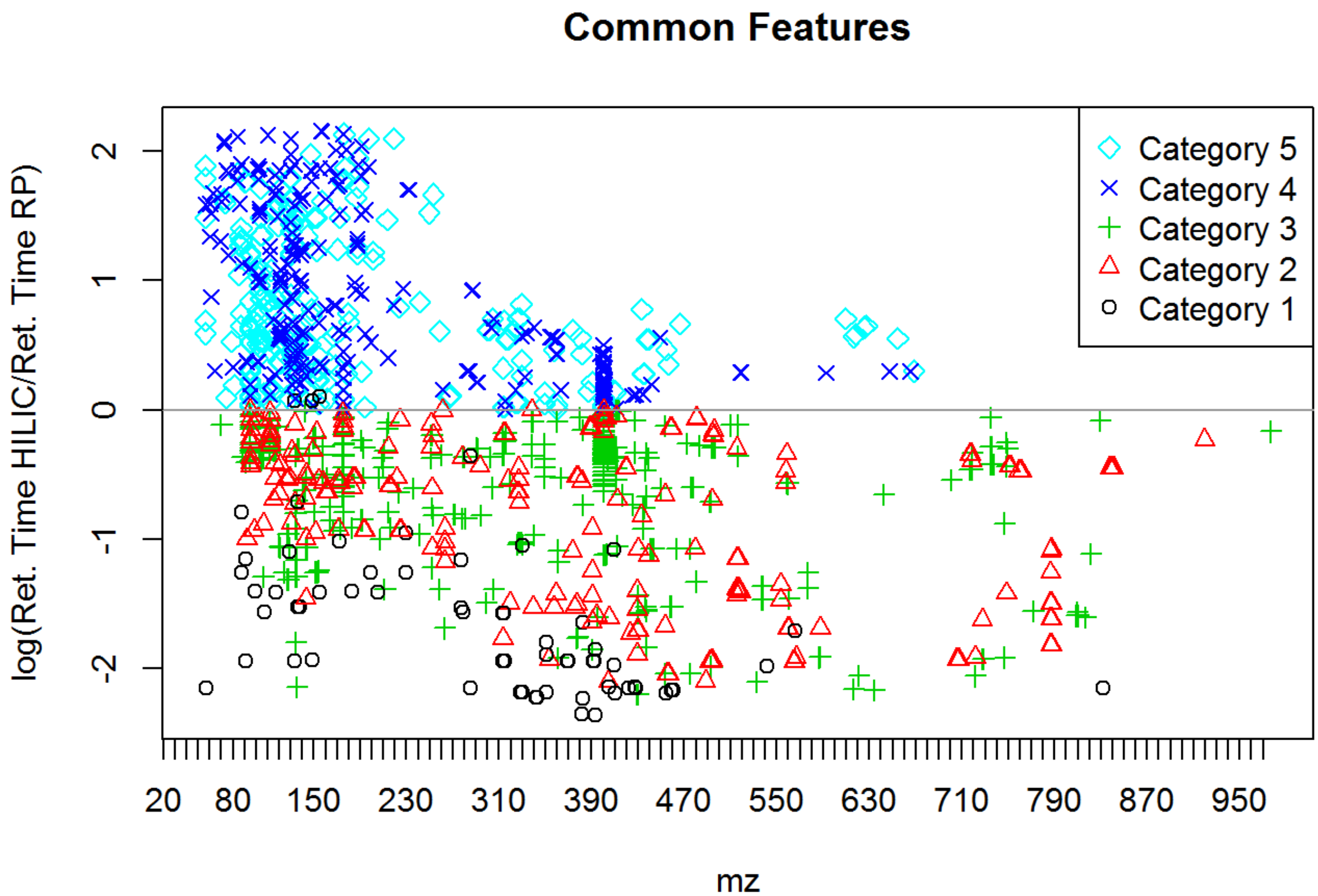
```
Verifying peak picking is done...

Detecting mass traces at 17.6 ppm ...
% finished: 0 10 20 30 40 50 60 70 80 90 100
2975 m/z ROI's.

Detecting chromatographic peaks ...
% finished: 0 10 20 30 40 50 60 70 80 90 100
2245  Peaks.
|=================================================================| 100%
Verifying peak picking on the other sample is done...
```

```
|=================================================================| 100%
Crossverifying of unique peaks is done...
```

```
|=================================================================| 100%
Crossverifying of unique peaks in the second sample is done...
```

```
  Step 7.Prioritization of common and uncommon peaks
  prioritized_output<-prioritization(output)
```

```
In total 1072 common features distributed in 5 caregories
Category 5: 246
Category 4: 242
Category 3: 329
Category 2: 191
Category 1: 64
Unique peaks in HILIC 394
Unique peaks in RP 1400
```




```
Step 8.Plot EICs
writeoutput(prioritized_output,
            sampleRP=sub_RP,
            sampleHILIC=sub_HILIC,
            type="o")
```

```
Creating pdf file with 448 EICs of unique peaks in HILIC
|==================================================================| 100%
Waiting...Writing pdf file
```

```
Creating pdf file with 1524 EICs of unique peaks in RP
|==================================================================| 100%
Waiting...Writing pdf file
```

```
Creating common EICs Category 5
|==================================================================| 100%
Waiting...Writing pdf file
```